

INCIDENsly webApp

Frontend generated with AI and integrated with a real Laravel REST API.

IT Academy
Sprint 5
Orlando Rojas Oyarzún

Documentation

This document presents the development process of a full-stack application, focusing on the use of AI to generate a functional frontend for a REST API.

The backend was built with Laravel and documented using Scribe, while the frontend was generated and refined using AI.

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What is INCIDENsly webApp?

Project overview

INCIDENsly *webApp*

INCIDENsly *webAPP* is an incident management system designed to streamline the reporting, tracking, and resolution of technical issues within an organization.

Technologies used

Stack

- > Frontend: React 19 + Vite, Tailwind CSS, React Router
- > Backend: Laravel 11 (PHP), RESTful API architecture
Authentication: Laravel Passport (OAuth2)
- > Deployment: Railway (backend) + Vercel (frontend)

Core features

- > User authentication (login/register)
- > Create, view, and manage incidents
- > Role-based dashboard (user vs admin)
- > Comments system for collaboration

Why MiniMax 2.7?

AI model selection

MiniMax 2.7

MiniMax 2.7 was selected due to its balance between cost, accessibility, and strong frontend code generation capabilities.

Primary focus

- > The AI was tasked exclusively with frontend development, including React component architecture, Tailwind styling, and UI/UX patterns.
- > It doesn't use tokens as others, it uses 'requests'. The amount of petitions you send it's measures by how may times you press 'enter', rather than counting tokens.

Criteria

- > Code generation capability for modern JavaScript frameworks
- > Availability and ease of access
- > Context window suitable for React component understanding. It has a short context windows, but I prefer this as I don't like to persist on AI sessions,
- > Multilingual support (helpful for Spanish-English translation)

Why not other options?

- > ChatGPT/Claude: Excellent options but higher cost barrier.
- > GitHub Copilot: Great for autocomplete but less interactive for learning
- > MiniMax 2.7: Provided a good balance of capabilities with accessible pricing, 10€

Key Conversations with MiniMax 2.7

Interaction with the AI

Initial prompt

>

“Crea un proyecto React 19 con Vite para mi aplicación de manejo de incidencias: Se trata de una dashboard CRUD creada con Laravel para gestionar incidencias. Utiliza una estética retro IBM Plex / E-Reader tipo Elnk. “

>

Utiliza una paleta monocromática de alto contraste (blanco y negro con sutiles grises), bordes nítidos y las fuentes IBM Plex Sans e IBM Plex Mono para lograr ese aspecto de "papel digital" o hoja de datos táctil. Basate en las referencias visuales que adjunto como forma de brainstorming

Visual references

>

신,
렇게
꾸세요.

#5F5F5F
기본적인 소프트 그레이

#FAF9F6
기본적인 소프트 그레이

#2F5F5F
블루 그레이 → IT/금융에 잘 맞음

#EDED
무난하고 균형 잡힌 그레이

14:41

Aeonik Fono

19.0°C

Heating up to desired temperature....

68%

RADIATORS

WATER 67°C

PRESSURE 1.7bar

TIMER ON OFF

PANELS

CAPACITY 320kWh

WEEKLY +1.6%

TOTAL YELD 190kWh

+ADD

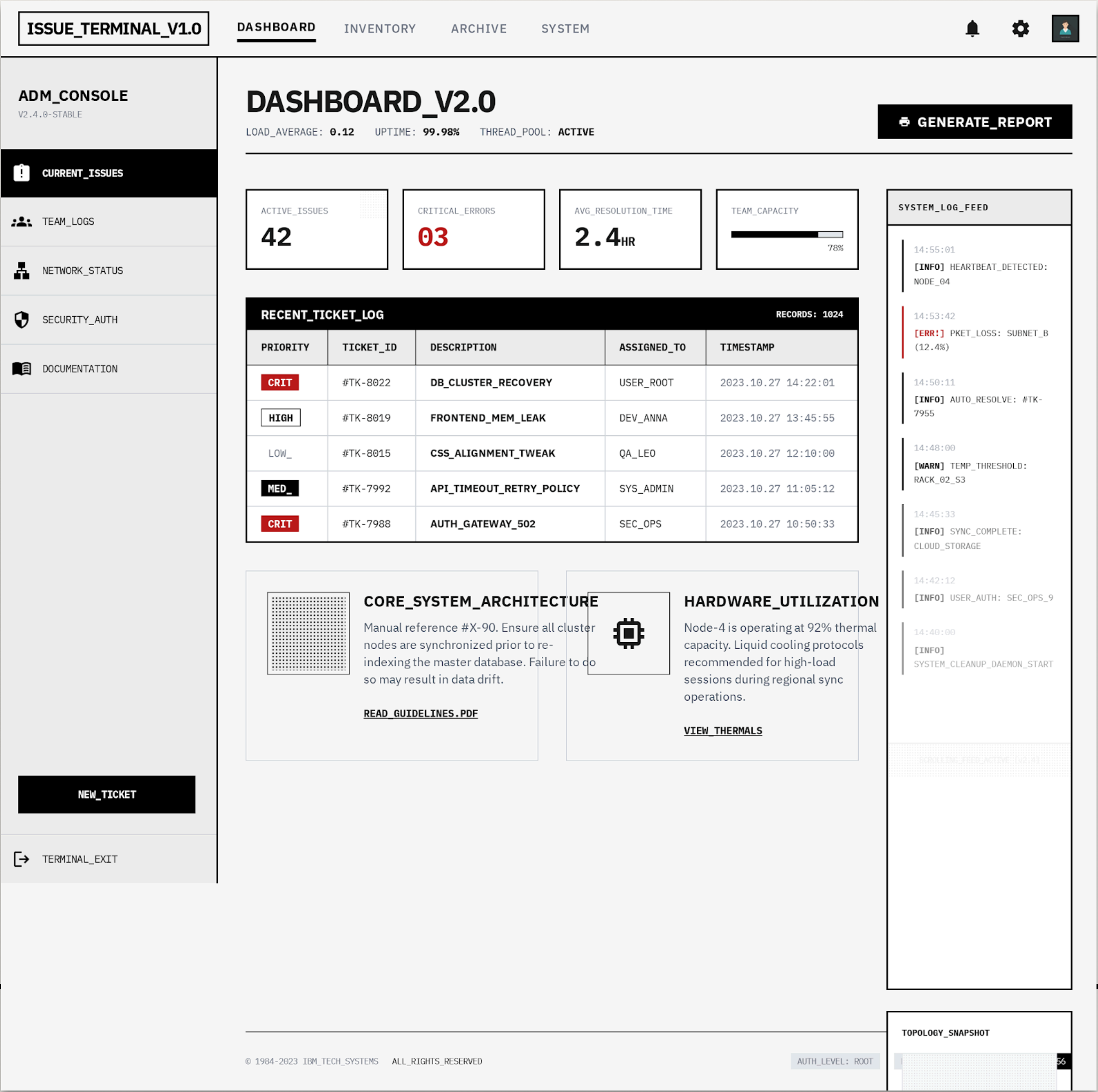
WEEKDAYS 18°C 6AM-10AM 5PM-11:30PM OFF

DAY OFF 19°C 9AM-1PM 5PM-11:30PM ON

Starting point

This is already a solid staring point, it has a lot of noise but in my experience, this would work for fine tuning.

I will always recommend to attach some visual references to frame a bit the imagination of the AI



Middle Point

NUEVA INCIDENCIA

TÍTULO

EJ: FALLO EN PASARELA DE PAGO

DESCRIPCIÓN

DESCRIBE EL PROBLEMA TÉCNICO...

TAGS

backend urgente bug

Separa con espacios.

ESTADO

Abierta

PRIORIDAD

Media

CANCELAR

CREAR INCIDENCIA

Create a new ...

“Ahora crea las vistas para cuando el usuario ve una incidencia, cuando crea una incidencia y cuando la edita, esto debe ser implementando con un modal que se despegue del fondo dejando un poco mas opaco para darle un poco de profundidad. Pon énfasis en los drop down menus de Prioridad y Estado que deben ser, para prioridad: High - Medium - Low para estado: open, in progress y closed. Es importante que mantengas la coherencia visual con el resto del sistema.”

INC-010

PRIORITY_A

DESCRIPTION

"No puedo acceder al sistema, me dice que la contraseña expiró"

REPORTER

Salem

TAGS

#URGENTE

#LOGIN

CURRENT STATUS

PRIORITY

ALTA

STATE

ABIERTA

ACTIVITY_LOG

DUNGEON GOBLIN

2024-05-24 09:12:04

Checking LDAP synchronization for user: SALEM. The password rotation policy was triggered 48 hours ago.

SALEM

2024-05-24 09:15:30

Can we bypass the lock? I need to finalize the quarterly report before 12:00.

POST_COMMENT

MESSAGE_BODY

Añadir comentario...

NOTIFY_STAKEHOLDERS

COMENTAR

QUICK_ACTIONS

ASSIGN_TO_ME

MARK_AS_RESOLVED

ESCALATE_CRITICAL

ID_SYSTEM_V4.2.1-SECURED

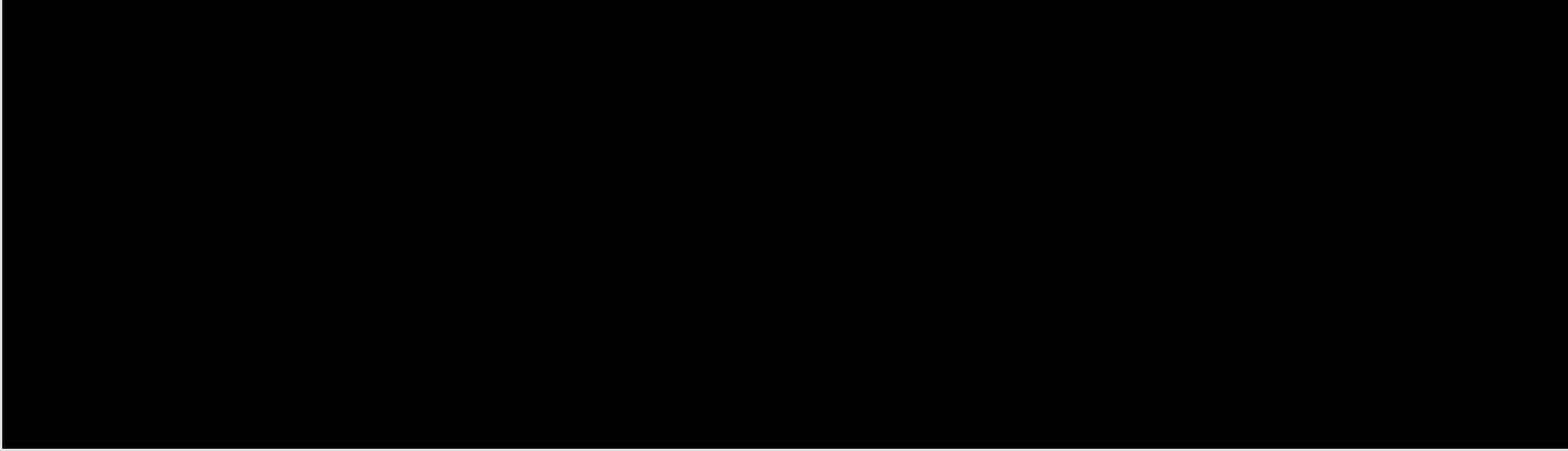
Modal purposes

“El modal debe incluir: una cabecera, puedes adornarla con el número de la incidencia y un botón de cierre. Debe contener un detalle de la incidencia: un texto destacado con la incidencia reportada. Información del Reportero y Tags: Visualización clara de quién reportó el problema y etiquetas asociadas. Recuerda también que la app tiene la opción de comentarios anidados para interacción entre usuarios: Hilo de Comentarios: Un historial de actividad con estilo de terminal, incluyendo avatares y marcas de tiempo, para facilitar el seguimiento.”

Last steps

“Con todo lo que haz diseñado, ahora nesitamos pasar a la implementacion React con componentes modulares. Este debe ser el paso mas importante en el proceso donde vamos a Modularizar las 1397 lineas de codigo que me haz pasado (jeje) en componentes bien definidos y separados. Recuerda utilizar componenetes individuales al estilo de React con arquitectura limpia y reutilizable.”





Setting connections

Backend <-> Frontend

Requirements for prompting connection

> API Service (src/services/api.js)

- Axios instance configured with a baseURL pointing to /api/v1.
- Request Interceptor: Automatically injects the JWT token from localStorage into the Authorization: Bearer header.
- Response Interceptor: Centralized error handling:
 - 401: Clears the token and redirects to login.
 - 403: Logs permission errors.
 - 5xx: Logs server errors.
 - Handles network errors with no response.

> Route Constants (src/utils/constants.js)

- Defines all API endpoints (/incidences, /metrics, /login, etc.).
- Prevents hardcoding strings within components.

> Custom Hooks (src/hooks/useMetrics.js)

- Manage loading and error states.
- Consume the API service to fetch and manage data.

Setting up

Axios connection

Crea un sistema de conexión con backend para React:

1. Servicio API con axios en `src/services/api.js`:
 - URL base desde VITE_API_BASE_URL + `/api/v1`
 - Interceptor que lea token del localStorage y lo adjunte como Bearer en Authorization
 - Interceptor que maneje errores 401 (limpia token), 403, 5xx y errores de red
2. Constantes en `src/utils/constants.js`:
 - Routes: incidences, my-incidences, metrics, me, login, logout, register, users
 - Priorities: high, medium, low
 - Statuses: open, in_progress, closed
3. Hook useMetrics en `src/hooks/useMetrics.js`:
 - Obtiene stats de /metrics: total, by_priority.high, by_status.open/in_progress/closed
 - Estados: stats, loading, fetchStats

■ Plan · MiniMax-M2.7 · 5.5s

Prompting . . .

```
Crea un servicio API centralizado para React usando axios que se conecte a un backend mediante una URL base configurable desde variables de entorno (VITE_API_BASE_URL) apuntando a /api/v1. El servicio debe incluir un interceptor de requests que lea el token JWT del localStorage y lo adjunte en el header Authorization como Bearer token, y un interceptor de responses que maneje errores comunes como 401 (limpia token), 403, errores 5xx y errores de red, siempre rechazando la promesa con el error. Además, crea un archivo de constantes con las rutas del API (incidences, my-incidences, metrics, me, login, logout, register, users), prioridades (high, medium, low) y estatus (open, in_progress, closed). Finalmente, implementa un hook personalizado useMetrics que consuma el endpoint /metrics, maneje estados de carga y error, y extraiga las estadísticas de total, prioridad alta, y estados open, in_progress y closed desde la respuesta data.data.
```

■ Plan · MiniMax-M2.7 · 6.6s

Personal reflections &

Knowledge notes

Limitations and strong points

First thing

> It's okay, It's very helpful for an initial approach. As someone with no prior design experience, it provided a good opportunity to start getting familiar with React

Second thing

> A 'must' for me is using AI to generate detailed academic documentation, including clear titles and references on what to edit. It has been incredibly useful—so much so that it's become a habit that encourages me to dive deeper into documentation whenever I hit a wall.

Third thing

> I think the process is quite slow and can become a bit annoying ... even, frustrating. For example, if you're trying to figure out tiny details—like an icon—and you ask AI to edit it, it's very likely it won't understand what you have in mind. Perfecting small details with AI is tedious; it's much better to 'sew' them by hand. And we all know ... small details makes the differences.

Fourth thing and last

> For me, it's a bit of a lifesaver when it comes to frontend, because I've always considered myself very weak with design aspects and anything related. I still try, but I admit this comes in very handy.

Additional information

Links & resources

Resources to share

REST API

> https://github.com/malmental/S5_API_REST

Frontend

> https://github.com/malmental/S5_API_frontend

API Docs

> <https://incidensly-webapp-production.up.railway.app/docs/>

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